

Dispersal and habitat connectivity in complex heterogeneous landscapes: an analysis with a GIS-based random walk model

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Schippers, P., Verboom, J., Knaapen, J. P. and van Apeldoorn, R. C. 1996. Dispersal and habitat connectivity in complex heterogeneous landscapes: an analysis with a GIS-based random walk model. – *Ecography* 19: 97–106.

A grid-based random walk model has been developed to simulate animal dispersal, taking landscape heterogeneity and linear barriers such as roads and rivers into account. The model can be used to estimate connectivity, and has been parameterized for the badger in the central Netherlands. The importance of key parameters was evaluated by means of sensitivity analysis. Results agree with field observations, and give interesting insight into the isolation of populations and potential populations. The model can be applied to obtain knowledge about dispersal processes in complex landscapes.

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Nowadays, in man dominated regions, many animal populations are relatively small and isolated. If these populations contain few individuals, the probability of local extinction is high (Richter-Dyn and Goel 1972). Dispersal is a key process in determining the survival of these so-called metapopulations, sets of local populations connected by inter-patch dispersal (Levins 1970, Opdam 1988, Verboom et al. 1993). The dispersal depends on the connectivity of the landscape (Fahrig and Merriam 1985, Apeldoorn et al. 1992). In order to design conservation strategies for endangered species, it is crucial to obtain insights into the relation between dispersal and landscape characteristics. Models can be used to describe dispersal and estimate connectivity. There are three different approaches that can be distinguished: distance based models, diffusion-like models and those based upon the random walk concept.

Most authors have used simple distance-based models to predict the connectivity between populations.

Waser (1985), Buechner (1987) and Miller and Carrol (1989) worked with geometric models to estimate the probability of dispersing animals reaching the next population. Exponential methods were used by Caley (1991). Verboom et al. (1991b) distinguished three connectivity classes based on distance. These models consider the landscape to be homogeneous. Most landscapes, however, have heterogeneous components such as different types of vegetation or barriers (roads, rivers) which can have a marked effect on animal dispersal.

Reeves and Usher (1989) simulated the spread of the coypu in the U.K. using a grid based model. The grid information was derived by map interpretation. Dispersal of the animals was simulated using a diffusion-like algorithm where animal flow between the grid cells was proportional to the density difference between the cells. Landscape heterogeneity was introduced by estimating the diffusion rate from the configuration of landscape

Accepted 31 May 1995

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ISSN 0906-7590

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elements in the grid cell. The grid size was 10 by 10 km and the dispersal front moved one grid cell per time step.

Knaapen et al. (1992) developed the model 'DISPERS' for planning purposes. This approach took barriers and landscape heterogeneity into account, the grid size was 1 km², and a Geographical Information System (GIS) was used to provide the model with landscape data. The algorithm was based on landscape resistance, which means that a certain landscape can stop the dispersal front for several time steps. Relative accessibility, was expressed as the number of time steps the model needed to reach certain grid cells. Because the time steps did not represent real time, the results of this model were only meaningful in a comparative way.

Animal movement was often simulated as a random walk process in a grid landscape (Johnson et al. 1992). Most of these simulation experiments were done in simple heterogeneous landscapes with at random dispersal habitat and non-accessible habitat (for a review see Johnson et al. 1992).

The models described above cannot be used to estimate the probability of a dispersing animal reaching another population in complex heterogeneous landscapes. The distance-based and random walk models ignore the underlying heterogeneity of the landscape. On the other hand, the diffusion-like algorithms cannot capture the complex behaviour of discrete individuals. Our objective in this paper is to obtain more insight into dispersal processes in complex heterogeneous landscapes. Therefore we derive and evaluate a new model, GRIDWALK, which estimates the connectivity of the landscape between existing (occupied) or potential (extinct or otherwise unoccupied) populations. The model can be used as a tool in landscape ecology and in modelling metapopulation dynamics. Here, the use of the model is illustrated using the European badger *Meles meles* in the central part of The Netherlands as a case study.

Model

Landscape and populations

In order to determine the connectivity between units of a metapopulation one must locate these units in the region under consideration. At any one time, some of these units of suitable habitat will be occupied (local populations) and others will be empty (unoccupied patches). In addition, the inter-patch landscape needs to be defined in terms of parameters that are relevant to animal dispersal.

The badger population in the central part of The Netherlands is chosen as an example to demonstrate the model. The population in this region has a metapopulation structure (Lankester et al. 1991), and badger habitat is isolated by rural and municipal areas and

Table 1. Badger regions and population reference numbers in the central part of The Netherlands (populations correspond to Fig. 1, potential populations in bold).

Area	Population number
Dunes	11
Western Utrecht	21, 22
Utrechtse Heuvelrug	31, 32, 33, 34, 35, 36, 37, 38
Gelderse Vallei	41, 42
Northern Veluwe	51, 52, 53, 54, 55, 56, 57
Southern Veluwe	61, 62, 63, 64, 65, 66, 67, 68
Eastern Veluwe	71, 72, 73, 74, 75, 76

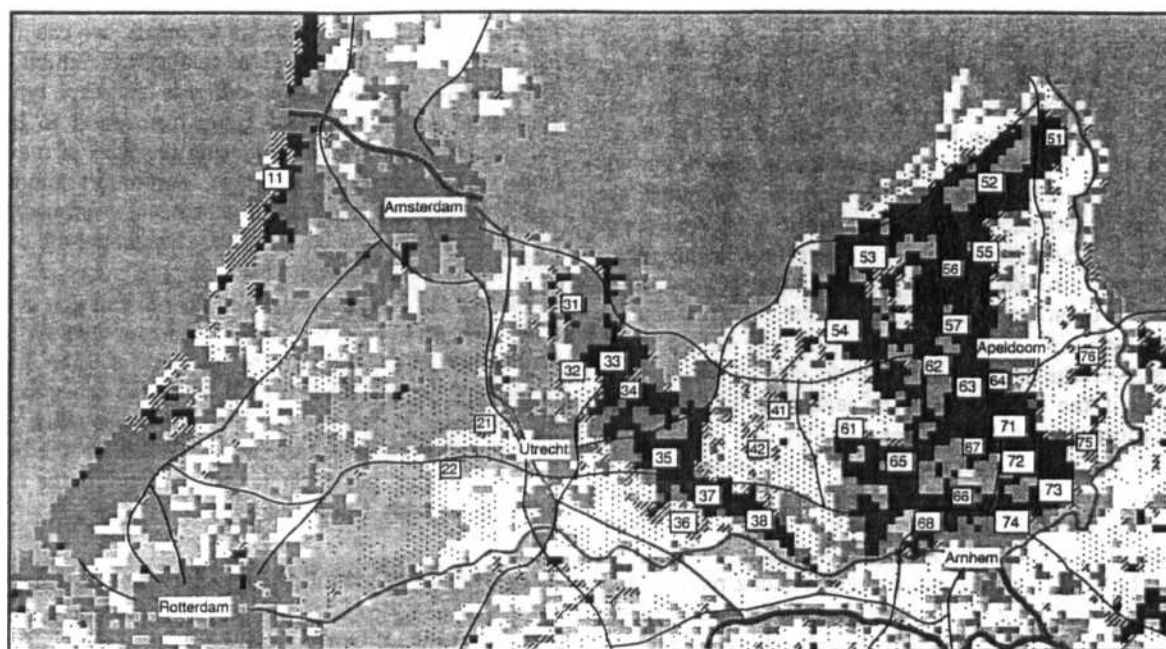
crossed by linear elements such as motorways and rivers. The borders of this area are formed by natural barriers, in the north the North Sea Canal and the IJsselmeer, in the east the river IJssel, in the south the river Lek and in the west the North Sea.

A population was defined as a number of inhabited setts, situated close to one another, and not separated from each other by major barriers. We distinguished three carrying capacity classes: two, four and eight setts. Each sett is located in one grid cell so a population of maximally 8 setts covers an area of 8 km². Twenty existing populations were marked out based upon the data of Wiertz and Vink (1983) and the landscape characteristics (Table 1). Fourteen potential populations were defined. These populations were projected in areas which resemble the natural habitat of the badger in terms of vegetation coverage (high), ground-water level (low) and soil type (sandy) (Fig. 1, Table 1). Throughout this paper, we use the term 'population' both when referring to existing populations, and to potential populations (suitable habitat patches).

The landscape surrounding the populations can be divided into several types that are relevant to the dispersing animals. These landscape types can be translated into movement and survival parameters which determine model behaviour. We distinguished 9 landscape types (Table 2) which are assumed to be relevant to the badger. These types differ in vegetation structure, food availability, and potential for excavating setts. For practical purposes the landscape was described as a square grid. We chose to use a cell size of one square km. The LKN (landscape ecological mapping of The Netherlands) files provided the information for creating the landscape maps on the square km level. If a grid cell consisted of more landscape types, the type which covered the largest area represents the cell.

Introduction to the algorithm

The random walk concept was applied in order to allow habitat preference and linear barriers to function stochastically: the behaviour of individual animals was simulated as a spatial stochastic process in which differences in landscape quality determined the probabilities



LANDSCAPES

	Woodland		Wet pastures with hedges
	Pastures with woodland		Wet pastures
	Heather		Arable land
	Dry pastures with hedges		Built-up area
	Dry pastures		Water / outside study area

BARRIERS

	Highway
	River
	Main canal

POPULATION NR. (MAX. SIZE)

	71 (8 Setts)
	35 (4 Setts)
	56 (2 Setts)

Fig. 1. Map of badger populations, landscapes and barriers in the central part of The Netherlands. The maximum size of populations is indicated by the size of the squares, the numbers are identification codes of the populations. One cell is 1 km². For further explanation see legend.

of different movement choices, and in which chance determined the actual path realized. In order to model dispersal behaviour mechanistically, the following components of dispersal were distinguished: movement within one landscape type, movement between different landscape types, mortality, interaction of the animals with linear barriers and finally the connectivity between the populations. These components are described in more detail in the following sections.

Movement within one landscape type

The velocity of an animal can be expressed in the reverse form as the 'residence time' in a grid cell. The longer an animal stays in a cell the slower it moves through a certain type of landscape. Dispersal speed was simulated by varying the residence time determined by the landscape type. We assumed that the better the quality, the longer the animal will stay.

Table 2. Estimated landscape quality and residence time values for dispersing badgers based upon sett potential and vegetation coverage for the use in the model GRIDWALK. Values used for sensitivity analysis of the model in brackets (low-high).

Landscape	Cover	Sett potential	Quality	Residence time (hour)
Water	-	-	0 (0)	0 (0)
Built-up area	-	-	0 (0)	0 (0)
Wet pastures	-	-	5 (5)	5 (2.5-10)
Wet pastures with hedges	±	-	15 (10-25)	5 (2.5-10)
Dry pastures	-	±	15 (10-25)	1 (0.5-2)
Dry pastures with hedges	±	±	25 (10-60)	1 (0.5-2)
Pastures with woodland	+	++	100 (20-400)	5 (2.5-10)
Heather	±	-	15 (10-25)	1 (0.5-2)
Woodlands	++	+	100 (20-400)	5 (2.5-10)
Arable land	±	±	25 (10-60)	2 (1-4)

Because there is no knowledge about the dispersal speed of badgers, we use knowledge about movement patterns of resident badgers as a reference for dispersal. Using walking patterns described by Kruuk (1989), it can be calculated that the speed of badgers between food patches within their territory varies from 0.76–3.5 km h⁻¹. The walking pattern within a square km will not be in a straight line. We therefore estimated the average residence time of a badger in a square km cell with a landscape type of no interest to be one hour. Badgers are known to prefer woody landscapes to totally open landscapes. Furthermore, a badger may be interested in the food availability and the sett potentials of a certain type of landscape. In general, we assumed that if the badger finds the landscape in a grid cell favourable, it will stay longer (Table 2).

If an animal leaves the cell, it must choose a direction. The probability of choosing a new direction was assumed to be dependent upon the old one. This means that the animal has a memory for direction (Pulliam et al. 1992), the result is more straight than a pure random walk. Movement in the same direction was thus judged to have a higher probability than directions perpendicular to the last movement. Returning to the old cell had probability zero. The probabilities of choosing a direction in a homogeneous situation were chosen to be: old direction: 0.50, perpendicular to the old direction: 0.25 and reverse to the old direction: 0.0.

The activity period of a badger on dispersal was estimated to be five hours per day. This was based upon the normal activity of badgers in a home range (Neal 1986). The maximum time the badgers were allowed to walk was four years.

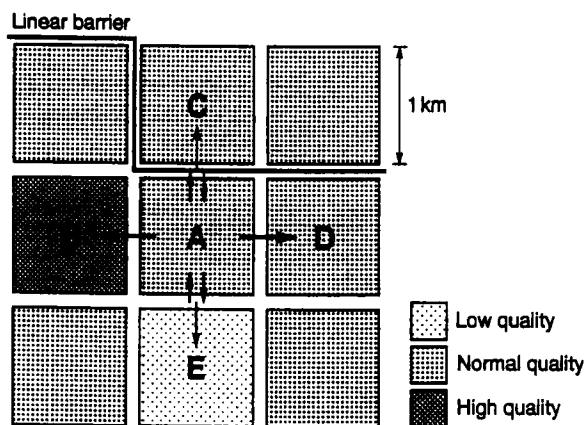


Fig. 2. Possible movements of a simulated animal from cell A in a grid landscape. Animals which move from A → E have a probability to reflect this direction due to the low landscape quality. If an animal moves in the C direction it will meet a linear barrier (for instance a motorway) which has repellent properties and can cause geographical isolation.

Movement between different landscape types

Animals prefer certain habitats to others, we call this factor 'landscape quality'. This quality can inhibit or facilitate movements between different landscape types; an animal will often avoid going from good to bad habitats. If an individual, following the rules of movement within one landscape type, encounters another landscape type, it must decide whether it continues its movement to the new landscape or stays in the old one. This choice is dependent upon the preference of the animal for the landscape types in the old and new cells. If the new grid cell is of a lower landscape quality, the probability of staying in the old cell will be high (movement from A to E Fig. 2). If on the other hand, the landscape in the new cell is of a higher quality, the animal will always continue its movement to the new cell. The animal has a chance to continue to the new cell (Fig. 2) with a probability of:

$$P(\text{move}) = Q(n)/Q(s) \quad \text{if } Q(n) \leq Q(s)$$

$$P(\text{move}) = 1.0 \quad \text{if } Q(n) > Q(s)$$

Where $Q(n)$ = quality of the new cell and $Q(s)$ = quality of the source (old) cell.

A badger on dispersal must find another population or a new area to settle. Therefore, sett potential was assumed to influence landscape quality. In The Netherlands, setts were usually located on dry spots with a woody vegetation coverage. In the research area most woods were on dry sandy soil which is suitable for digging setts. The best landscape types were assigned to have quality of 100, other landscapes were rated lower with respect to their sett potential (Table 2). If a simulated animal refused to go to an area of low quality from a certain cell, it was not likely to try this direction a second time before trying out other directions. Surrounding cells which were rejected were thus no longer considered to be an alternative for the animal except when all four directions have been tried out.

Mortality

It is possible that the animal dies during its stay in a grid cell. Two components of mortality can be distinguished: natural dispersal mortality and 'additional grid mortality'. Natural dispersal mortality is the background mortality and is independent of the grid cell characteristics. Additional grid mortality is caused by the specific situation in a grid cell, and determined by factors such as lack of food or coverage, and the occurrence of roads.

For the badger, the probability of dying from natural causes was estimated to be 12.5% yr⁻¹ (Lanckester

1989). We estimated the natural mortality for a dispersing animal to be twice as high. Van der Zee et al. (1992) stated that the decline of the badgers in The Netherlands between 1960 and 1980 was closely related to the number of roads in the area, therefore the additional grid mortality was assumed to be proportional to road length and the calculation was based upon the assumption that the average survival time of a badger in a cell with one km of metalled road is one year. We derived this from the data of Berendse (1986) who demonstrated that traffic accidents were the main cause of death for the badger.

Linear barriers

Besides small roads which are easy to cross, major motorways, canals and rivers can be important barriers for dispersing animals. Their general properties are that their length is longer than the size of a cell but their width is much smaller. If a dispersing animal meets such a barrier there is the probability that it will not cross it, but if it does cross the barrier there is a chance that it dies. In the model these linear elements were thought to be located between the grid cells (Fig. 2). These barriers were characterized by two parameters, crossing mortality and barrier quality. Barrier quality was expressed in the same units as landscape quality. The model took the minimum value of the new cell and the linear barrier as the quality of the new cell. This new cell quality determined the chance of moving to the new cell (Fig. 2 movement from A to C). The chance of moving across a barrier was as follows:

$$P(\text{move}) = (\text{MIN}[Q(n), Q(b)] / Q(s)) \quad \text{if } \text{MIN}[Q(n), Q(b)] \leq Q(s)$$

$$P(\text{move}) = 1.0 \quad \text{if } \text{MIN}[Q(n), Q(b)] > Q(s)$$

where $P(\text{move})$ = probability of moving to the new cell. $\text{MIN}[a, b]$ is a function which takes the minimum value of a and b , $Q(n)$ = quality of the new cell, $Q(s)$ = quality of the source cell, $Q(b)$ = quality of the barrier between the source cell and the target cell. Only important barriers were taken into account (Table 3).

Barriers have repelling properties. In terms of the model, their quality must be lower than the minimum landscape quality (except impermeable landscape). On

Table 3. Estimated quality and crossing mortality values of linear barriers for the badger dispersal model. The values used for sensitivity analysis are in brackets.

Type	Quality	Mortality
Motorways	4 (2–8)	15% (7–30%)
Rivers (Lek, IJssel)	4	10%
Important canals	4	50%

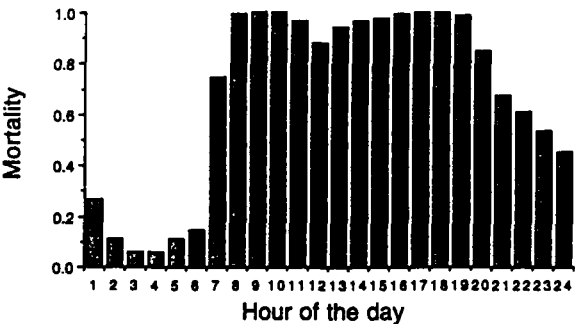


Fig. 3. Estimated mortality probability of a badger crossing a motorway assuming that it crosses the road perpendicularly with a speed of 3 km h⁻¹ and the motorway has a traffic intensity of 40 000 vehicles a day in both directions. The traffic intensity originates from measurements on the motorway between Utrecht and Apeldoorn where the road passes through the Veluwe area.

the other hand, no barrier is completely impermeable, so the quality in this case must be non zero.

The crossing mortality of a barrier was expressed as the probability of death per crossing event. For the badger, calculations of collision probabilities gave some insight into the chance of surviving a motorway crossing. Calculations showed that during the day a badger had little chance of surviving a crossing but at night there were some possibilities (Fig. 3). Because the period of activity of a badger is at night (Neal 1986) and the traffic intensity is then low, we estimated mortality while crossing a motorway to be 15%. Field observations show that badgers are able to cross rivers (Neal 1986). We assumed that the probability of a river crossing resulting in death was lower than that of crossing a motorway. Canals are more dangerous than rivers because of steep quays where it is often not possible for the badger to climb out. We therefore assumed the mortality value for crossing important canals to be 50% (Table 3).

Estimation of connectivity

A simulated individual, starting from a randomly selected sett of a source population moved according to the rules of the model determined by residence time, landscape quality, barrier location, and mortality, until it arrived in one of the setts of a target population or died on the way. Because of the stochastic character of the model, animals had different walking patterns (Fig. 4) and could arrive in various target populations. Arrivals and mortalities were registered. The connectivity between source and target population was estimated to be the fraction of animals starting in the source population which arrived in a certain target population. For an overview of all relations between the populations, each population was temporarily considered to be a source population while all the others were target populations.

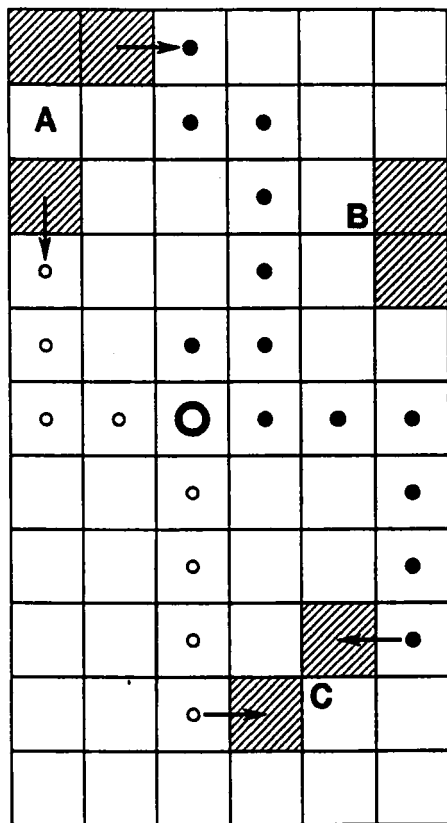


Fig. 4. Map of source (A) and target (B, C) populations, each grid cell of a population is one sett (shaded square). The dots represent walking patterns of two animals starting in population A and arriving in population C. Different walking patterns were caused by stochastic and landscape characteristics. The large dot indicates that this cell was passed by two different animals.

Sensitivity analysis

Method

Sensitivity to parameter changes was tested in order to evaluate the model. As an output parameter of the model, connectivity between population 61 and eight surrounding populations was used (Fig. 1). All other populations were still active as receptive (target) populations but were not evaluated. The analysis was done by changing one parameter or one group of parameters at a time. In general, values were chosen which were twice or half as high as in the reference situation (Table 1) except in the sensitivity analysis for quality. Here two alternatives were chosen, one with less differences between the landscape types and one where the differences were extended (Table 2).

To test how many released animals gave reproducible results, seven different runs were made with different numbers of animals (500, 1000, 5000, 10 000, 20 000, 50 000, 80 000) in the reference situation. A run with 10 000 animals was found to give reproducible accu-

racy. Results of the sensitivity analysis were compared with the reference run (Tables 2 and 3).

Results

Figure 5 shows the fraction of arrivals out of 80 000 animals started from population 61, arriving in the eight target populations (reference situation). More than half of the animals (54%) arrived in the nearest population 65. This was three times as much as the number of arrivals in population 62 which is isolated because of landscape characteristics (Fig. 1). Populations 41, 42 and 54 are isolated by landscape differences and motorways (Fig. 1). They have arrival values of c. 4%. Populations 35 and 38 were the most isolated ones; the animals must cross two motorways and pass at least one important change of landscape. The chance of reaching these populations from population 61 was c. 0.4%. Population 63 had a very low chance of arrival because of low quality heathland and villages between populations 61 and 63. Also, many animals were absorbed by populations 62 and 65 before they could reach 63. About 16% of the released animals arrived in populations other than the eight evaluated.

The effect of lowering the residence time (Fig. 6A) was that more animals arrive in far away populations. Increasing the residence time had the reverse effect. In general, high residence time increased grid mortality while barrier mortality decreased. Lowering the residence time had again the reverse effect.

Landscape quality appeared to be an important factor (Fig. 6B). The effect of reducing landscape quality differences, was that animals were no longer bound by the high quality forest area where the source population 61 was situated, therefore arrivals in popu-

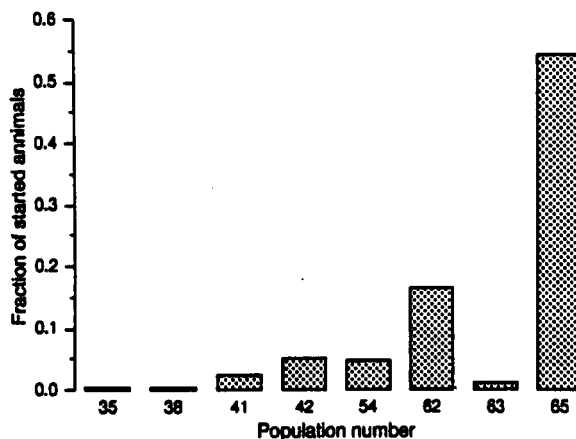


Fig. 5. Arrivals at eight populations, in the complex area surrounding population 61, as a fraction of 80 000 animals originating from population 61. The location of the populations is described see in Fig. 1 and Table 1.

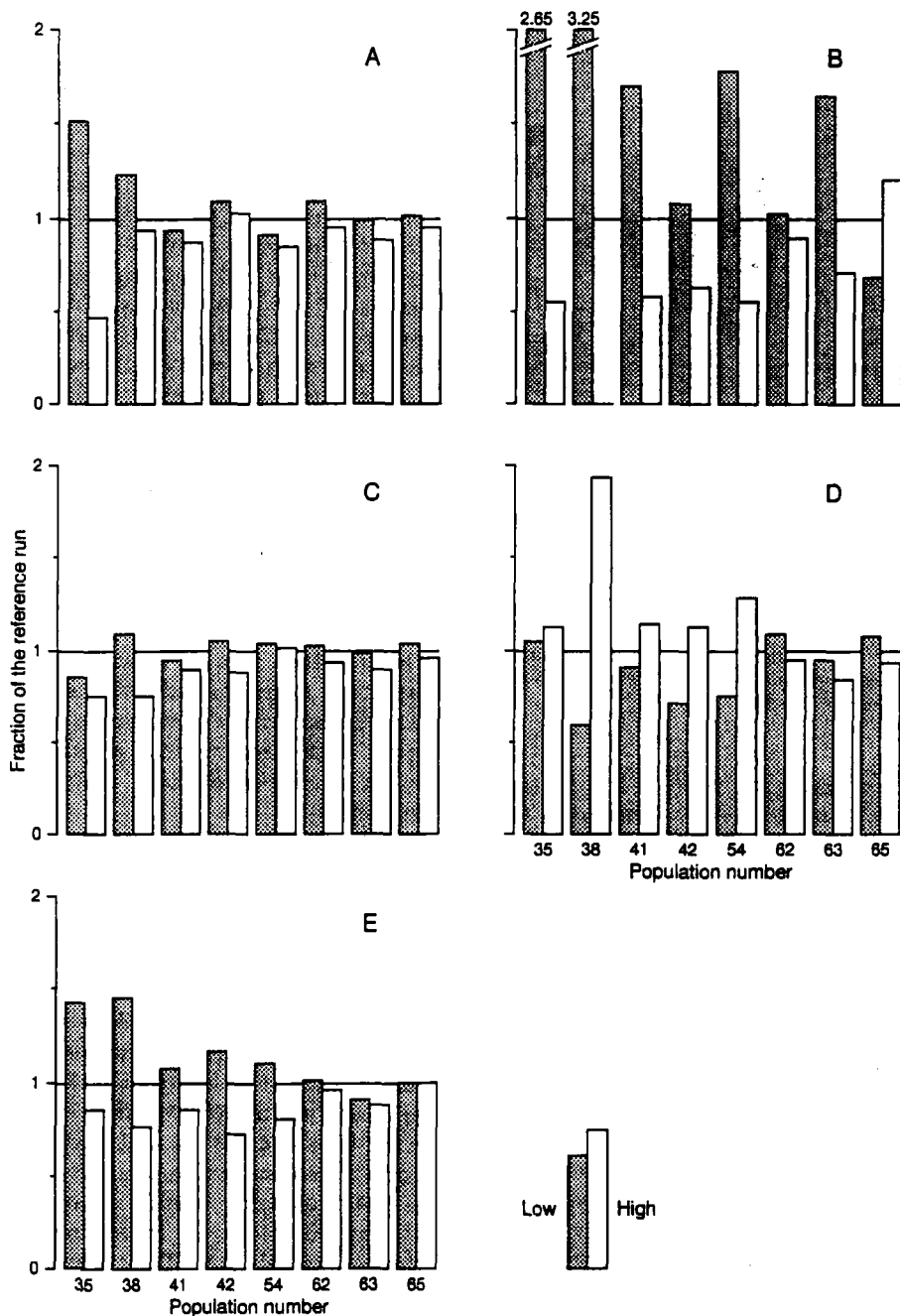


Fig. 6. Results of the sensitivity analysis. A: sensitivity to the residence time, B: sensitivity to landscape quality, C: sensitivity to grid mortality, D: sensitivity to barrier quality, E: sensitivity to barrier mortality. The vertical axis of all graphs represents the quotient between the fraction which arrived at the population as a result of the sensitivity analysis compared with the results of the standard run (Fig. 5). For an overview of population locations see Fig. 1. High and low refer to values used for the sensitivity analysis described in Table 2.

lation 65 decreased in favour of all the others. Populations 35 and 38 in particular take advantage of this change. An increase in quality differences had the reverse effect.

Figure 6C indicates that the model is robust for changes in grid mortality. The low mortality situation in particular caused few changes. The reason for this

is that animals which had less chance of dying live longer, which increased their chance of dying elsewhere due to other causes. For instance, lowering grid mortality leads to an increase in mortality caused by barriers.

Sensitivity to linear barriers is shown in Fig. 6D and 6E. The populations can be grouped into three differ-

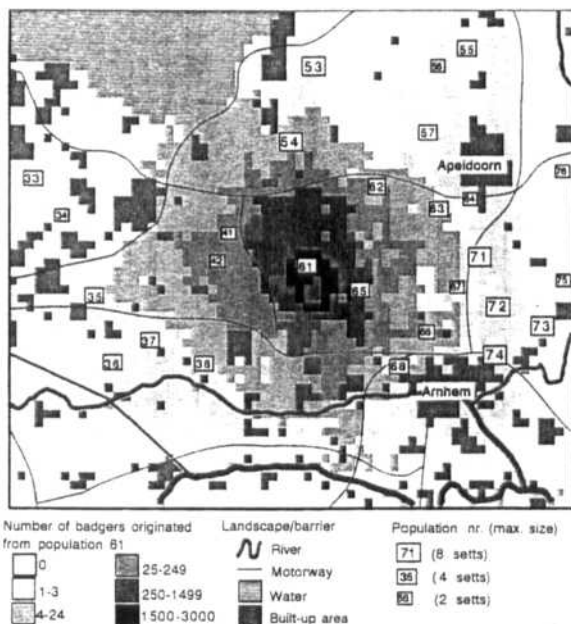


Fig. 7. Accumulated walking patterns of 10 000 simulated badgers started in population 61. One cell is 1 km². For further explanation see legend.

ent types, for explaining the results of Fig. 6E and 6F. Populations 62, 63 and 65 are not separated by motorways from the source population. Populations 54, 41, 42 and 35 are separated by one motorway, and most animals started in population 61 have to cross two motorways to reach population 38. Populations 65, 63 and 62 showed only slight differences compared to the reference. Population 38 was the extreme case while the other populations showed intermediate results. Decreasing motorway quality resulted in less crossings by animals which led to less mortality. The number of badgers which stayed on the source side of the road also increased, therefore the number of arrivals to the populations in this area increased. Another effect was that mortality decreased because fewer animals tried to cross the motorway. Increasing the quality had the reverse effect. The effects of changing level of mortality were almost the same as the causes of the quality change. Increasing mortality lowered the arrivals in all cases, decreasing mortality had the reverse effect.

An application of GRIDWALK

In a metapopulation context we are interested in the spatial relations between all the populations or potential populations. GRIDWALK can be used to gain insight into the total connectivity network of an area. The model can also locate inaccessible areas where animals have difficulty in dispersing.

Method

The population connectivity network was estimated by counting the number of animals arriving in a target population divided by the number of animals (5000) which started from the source population (=connectivity P). This was done for each population as a source population. This calculation produces a connectivity matrix which describes the relation between all the populations and potential populations based on arrivals of 170 000 animals. To evaluate dispersal patterns, a run with 10 000 individual animals was performed with population 61 as releasing point. A frequency map of accumulated walking patterns was made by counting the number of individual animals, that pass a cell in the grid landscape. Both simulations were done under reference conditions.

Results

The frequency map is shown in Fig. 7. Note that significant drops coincide with important landscape changes and barriers (compare Fig. 7 with Fig. 1). The spread to the eastern side of population 61 is inhibited by the absorption of many animals by the receptive populations.

Figure 8 describes the relations between all populations. This figure gives insight into the connectivity network, and the bottlenecks between groups of well-connected populations. The most isolated population is 11 (see also Fig. 1), which population is almost totally enclosed by a rural area. An animal needs to cross one motorway and walk at least 50 km to reach the nearest population. Populations 21 and 22 form a small cluster which is well isolated from population 32 by a canal and a motorway ($P = 0.0016$). Furthermore, a few lakes and the city of Utrecht make the connection virtually impossible. Animals from population 31 had a reasonable chance of reaching populations 32 and 33. However, in the reverse direction the chance was only 0.01. The cluster of populations 32, 33 and 34 was well isolated from the group consisting of populations 35, 36, 37, 38, 41 and 42 ($P = 0.037$) (southern Utrechtse Heuvelrug, Table 1). Another main bottleneck can be found between the latter group and the other populations which are situated in the well connected forest area called the Veluwe. The connection probability between the Veluwe area and the southern Utrechtse Heuvelrug was lower than 0.05. The isolation is caused by the landscape change from forest to open pastures, and one motorway. The Veluwe area is split into a northern and a southern part, the weak connection between these parts was caused by a motorway and a few large heathered areas.

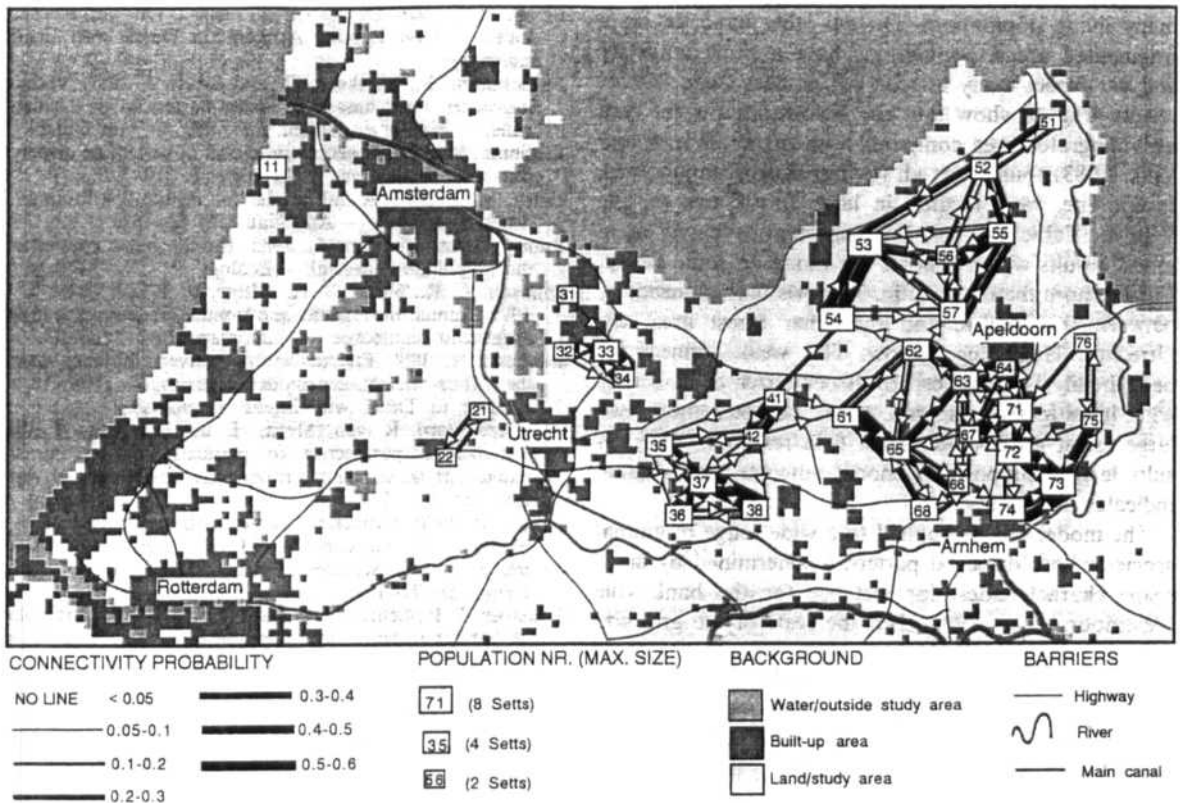


Fig. 8. Connectivity network between badger populations in the central part of The Netherlands as calculated by the model. The thickness of the connection lines is proportional to the connectivity between the populations. The direction for which the probability is valid is indicated by the arrow. One cell is 1 km².

Discussion

The advantages of a modelling approach are that available knowledge has to be structured, one is forced to estimate unknown parameters, and hypotheses have to be formulated about unknown mechanisms. Preliminary calculations can give quick insight into the consequences of assumptions and the sensitivity of parameters. Hence, a modelling approach can give an idea of the importance of gaps in knowledge and can structure future experimental research.

Although relatively simple, the model demands the quantification of quite a few parameters to describe the properties of landscape types and the response of the animals to them. Although an ordination of parameter values is usually simple (for example what is the order of preference of the landscape types), the data necessary for exact quantification are usually insufficient. For the badger, many parameters need to be chosen based upon common knowledge. We did a sensitivity analysis to evaluate these choices for the model outcome.

The sensitivity analysis shows that the model is especially sensitive to the response of the animal to land-

scape quality and barrier parameterization. This indicates that landscape and barrier configurations (especially highways) are probably important factors. Hence, more effort should be put into quantifying these parameters, and if we wish to propose management measures to improve the prospects for the badger, the measures must be related to these parameters.

Exact validation of the model for the badger is difficult if we consider that the estimated number of badgers performing long distance dispersal in the area is in the order of 10 yr⁻¹. This makes it almost impossible to estimate connectivity from experiments. One possibility would be to determine the genetic relation of existing populations. However, this is the outcome of a long-term process and would reflect the landscape of the past more than the present situation. Furthermore, connectivity between existing and potential populations cannot be determined in this way.

Another possibility is to compare the distribution of existing populations in the field with the connectivity network estimated by the model. We expect that a well-connected area has a lower probability of extinction because individual populations in such areas have a greater chance of recolonization. Here we expect

many living populations. On the other hand, in more fragmented areas, populations have a short longevity and we expect many unoccupied patches there. Model results (Fig. 8) show that the Veluwe area is the best well-integrated area compared with others. Wiertz and Vink (1983) found that all (living) badger populations, except one, were located in large forest area on the Veluwe (Table 1). This fits expectations based upon model results which validate the model to some extent. Furthermore the model (Fig. 8) shows that connectivity between the Veluwe and the other forest area, the Utrechtse Heuvelrug, is low. This weak connectivity had already been noticed by Broekhuizen et al. (1989) who investigated dispersal zones between these two large forest areas based upon field research. Their results largely support the model outcome which again indicates its validity.

The model can be applied to a wide range of animal species whose dispersal pattern is determined by landscape characteristics, for instance for the bank vole (Apeldoorn et al. 1992). But the scale of the grid size must be chosen according to the perception radius of the species. If the scale is too fine, the model animal obtains information about the neighboring cell, and a different algorithm is needed. If the scale is too coarse, relevant details about the landscape are lost.

The object of this paper is to obtain insight in dispersal processes and evaluate the model GRIDWALK as a tool for estimating habitat connectivity. Figures 7 and 8 show that the model can give insight into the dispersal pattern and reveal the connectivity network, which makes it possible to locate dispersal bottlenecks. Frequency maps from one source population can indicate where animals disperse to and where the risky spots are. These results, together with the flexibility of the model indicate that GRIDWALK can be an important tool for obtaining more knowledge on dispersal processes in complex landscapes. Combining results of GRIDWALK with metapopulation models developed by Lankester et al. (1991) and Verboom et al. (1991a) would give more realistic information about badger population dynamics in relation to the landscape.

Acknowledgements – The authors thank H. Meeuwssen, H. van Engen for their contribution to the discussion and their help creating some figures and further we thank H. Metz and J. van Groenendaal for their comments on the manuscript.

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